

Subarray Squares

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n elements, your task is to divide into k subarrays. The cost of each subarray is the square of the sum of the values in the subarray. What is the minimum total cost if you act optimally?

Input

The first input line has two integers n and k : the array elements and the number of subarrays. The array elements are numbered $1, 2, \dots, n$.

The second line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the minimum total cost.

Constraints

- $1 \leq k \leq n \leq 3000$
- $1 \leq x_i \leq 10^5$

Example

Input	Output
8 3 2 3 1 2 2 3 4 1	110

Explanation: An optimal solution is $[2, 3, 1]$, $[2, 2, 3]$, $[4, 1]$, whose cost is $(2 + 3 + 1)^2 + (2 + 2 + 3)^2 + (4 + 1)^2 = 110$.